NOT MEASUREMENT SENSITIVE
MIL-P-85856/37
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS MAGNESIUM ZIRCONATE, TYPE H, CLASS 1, COMPOSITION G

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

SiO_2	CaO	Al_2O_3 (1)	TiO_2 (1)	Fe_2O_3 (1)	MgO	ZrO_2	Others (1) (2)	
							Each	Total
1.5 max.	1.5 max.	1.0 max.	0.5 max.	0.5 max.	18.0 25.0	Bal	-	1.5 max.

(1) Determination not required for routine acceptance.

(2) Other elements total Include Al_2O_3, Fe_2O_3, and TiO_2.

Particle Size/Distribution

+200 Mesh (+75 µm)	1.0 max.
+270 Mesh (+53 µm)	7.0 max.
-325 Mesh (-45 µm)	55.0 min.

Sprayed Coating Properties

Bond Strength	17.3 MPa (2500 psi) min.
Hardness	Rockwell 15N, 75 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 10% max.

AMSC N/A FSC 3439

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimlted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:33Z
Check the source to verify that this is the current version before use.

Deposit Efficiency and Deposition Rate

70% min for N_2/H_2 @ 11.0 lb/hr
55% min for A/Hz @ 5.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	A50TF155, Class A
Pratt & Whitney Aircraft	PWA 1333
Rolls-Royce Ltd.	MSRR 9507/21

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/37)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-85856/37	POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:33Z
Check the source to verify that this is the current version before use.

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, were too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 53
Lakehurst, NJ 08733-5100

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 53
Lakehurst, NJ 08733-5100

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:33Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:33Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:33Z
Check the source to verify that this is the current version before use.